

AfyaPlus: Generative AI Operational Status

Engineering Decision Memorandum

To: CTO & Medical Director | From: AI Engineering (Observability) | 22 September 2026

Executive Summary

After six weeks in production with zero automated evaluation, drift monitoring, or API cost tracking, we built and ran all three observability layers this week. Headline finding: gpt-4o-mini is clinically safe for insurance verification and medication calculation, but it fails our clinical quality gate on emergency triage routing (Token F1 0.446 vs. 0.450 threshold). Simulated production traffic also shows statistically significant drift in patient input length and answer quality by month 2.

Quality Performance Breakdown

15 clinical questions (5 each: triage, insurance, medication) were scored against verified clinical references using BLEU, ROUGE-L, Token F1, and an LLM-as-judge (correctness, groundedness, relevance, helpfulness, overall alignment; 0-5 scale). gpt-4o passed the quality gate on all 3 features. gpt-4o-mini passed insurance verification and medication calculation, but narrowly failed the triage-routing gate on lexical overlap with the reference answer - a small but real miss on our most safety-critical feature. Judge correctness/groundedness scored 5/5 across the board in this run; we recommend a larger, harder judge rubric before fully trusting pass/fail at production scale.

Cost & Efficiency Analysis

At today's 75%/25% gpt-4o-mini/gpt-4o canary split, 30-day spend projects to \$29.30. Routing each feature to the cheapest model that clears its quality gate would save \$7.42/mo on insurance and \$4.76/mo on medication (\$12.18/mo combined) - but requires moving all triage traffic to gpt-4o, since gpt-4o-mini is not currently safe there (+\$33.82/mo). Net effect: \$29.30 -> \$50.95/mo, exceeding our \$35.00 monthly cap by ~46%. For context, fully human-reviewing triage volume (~31,415 requests/month) at a conservative 30 sec/request and a \$15/hr loaded clinician cost would run ~\$3,927/month - the AI path, even fully fixed, stays over 98% cheaper than manual review.

Systemic Operational Risks

Drift monitoring on simulated traffic (Evidently AI, Kolmogorov-Smirnov test, $p < 0.05$) already flags significant drift in patient input length and answer ROUGE-L by month 2, and in response latency by month 3. Simulated input length more than tripled (month 1: 20 tokens -> month 3: 65 tokens) as patient messages grew longer and noisier, while ROUGE-L against the clinical reference fell 40% (0.512 -> 0.305). Left unaddressed, this trend compounds the existing gpt-4o-mini quality-gate failure on triage and increases hallucination/misrouting risk on our most safety-critical feature.

Actionable Engineering Roadmap

- 1) Immediately stop routing triage-routing traffic to gpt-4o-mini; move 100% to gpt-4o (safety-critical, +\$33.82/mo).
- 2) Route insurance-verification and medication-calculation traffic fully to gpt-4o-mini - both clear the quality gate - to recover \$12.18/mo, partially offsetting (1).
- 3) Request a monthly budget cap increase to at least \$55 to accommodate safe routing, and re-run drift detection monthly - simulated patient input patterns already shift by month 2, not month 3, so a quarterly cadence would miss the earliest warning signs.